US 20250275207A1

(19) **United States**

(12) **Patent Application Publication**

(10) **Pub. No.: US 2025/0275207 A1**

(43) **Pub. Date: Aug. 28, 2025**

Chung et al.

(54) **HYBRID GAN AND BCD DEVICES USING HETEROEPITAXY ON SILICON**

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(21) Appl. No.: **18/589,545**

(22) Filed: **Feb. 28, 2024**

Publication Classification

(51) **Int. Cl.**

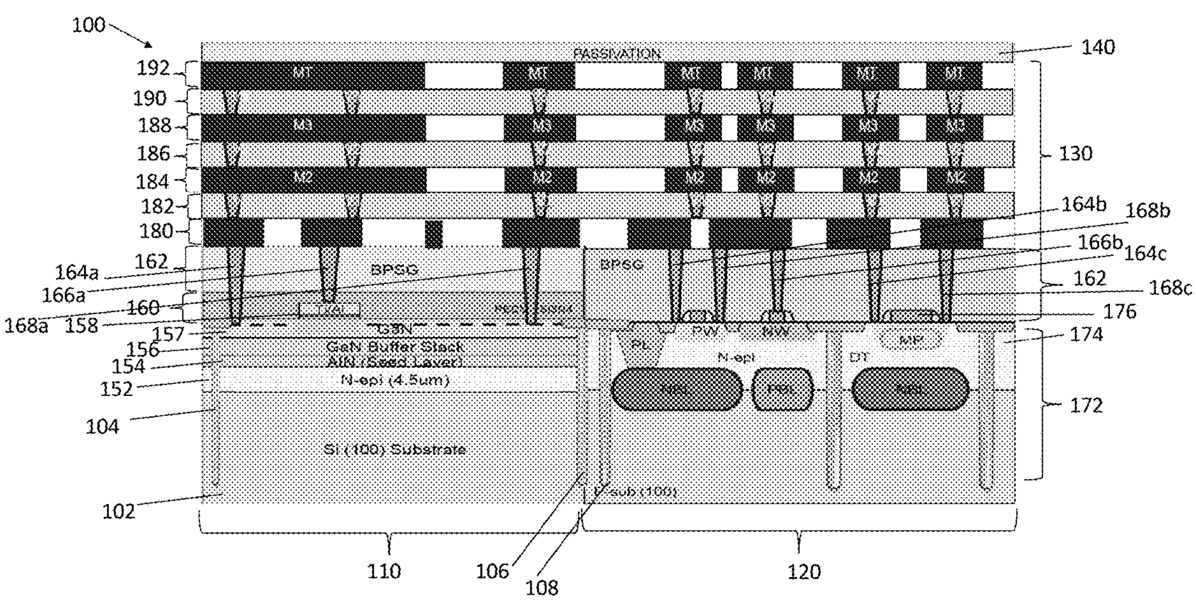
H01L 29/20	(2006.01)
H01L 21/02	(2006.01)
H01L 23/31	(2006.01)
H01L 27/092	(2006.01)
H01L 29/04	(2006.01)
H01L 29/66	(2006.01)
H01L 29/735	(2006.01)

(52) **U.S. Cl.**

CPC ... ***H10D 62/8503*** (2025.01); ***H01L 21/02422*** (2013.01); ***H01L 21/02433*** (2013.01); ***H01L 21/0254*** (2013.01); ***H01L 23/3171*** (2013.01); ***H10D 10/061*** (2025.01); ***H10D 10/60*** (2025.01); ***H10D 62/405*** (2025.01); ***H10D 84/85*** (2025.01)

(57) **ABSTRACT**

According to one aspect of the present disclosure, a semiconductor device includes a first substrate having a lattice structure, wherein the first substrate includes a gallium nitride (GaN) area adjacent to a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area. In some embodiments, the GaN area comprises one or more GaN device layers disposed on the first substrate. In some embodiments, the BCD area comprises one or more BCD device layers. In some embodiments, the first substrate comprises a silicon (100) lattice structure configuration. In some embodiments, the GaN devices layers comprise one or more GaN device layers having a cubic structure and one or more GaN device layers having a wurtzite structure.



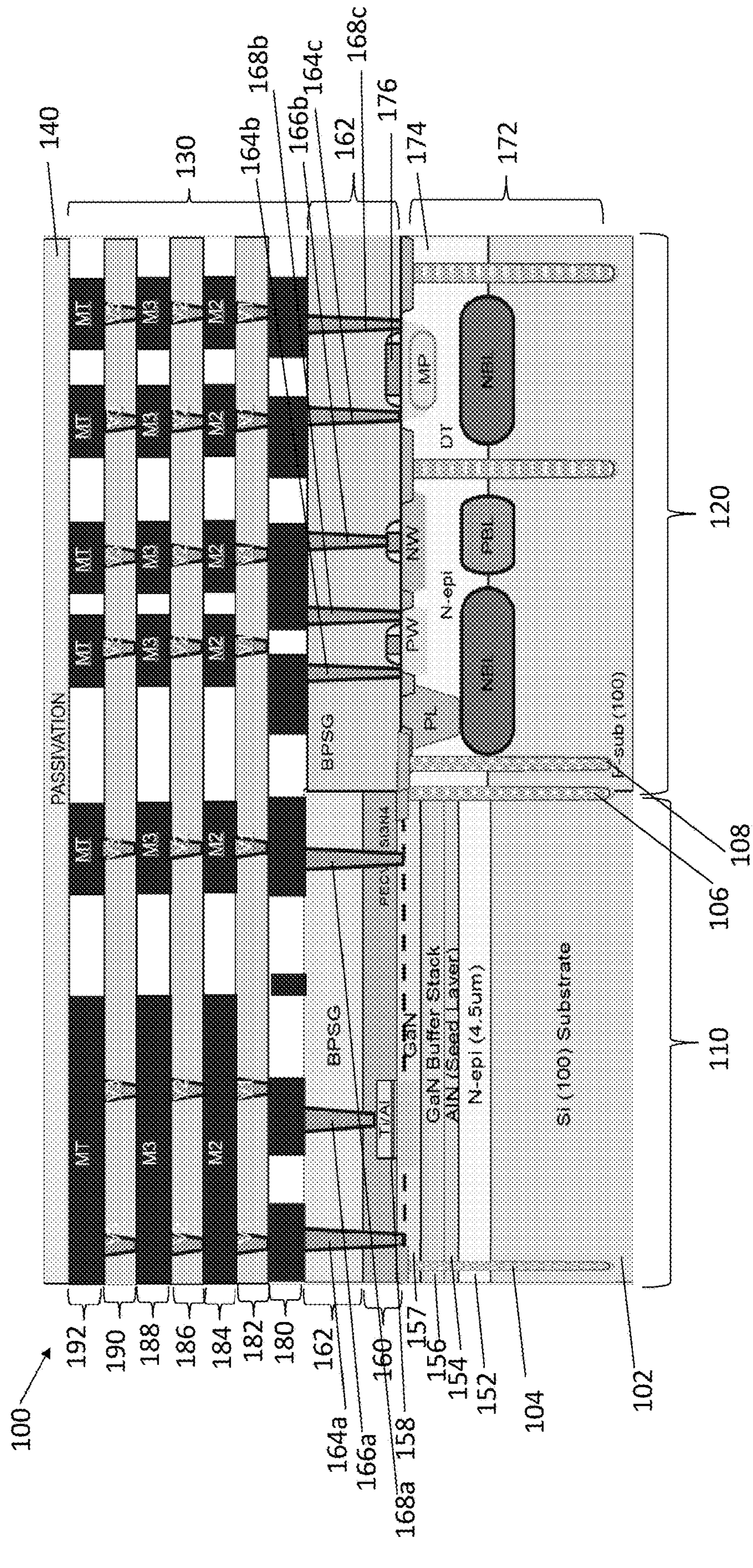

100
PASSIVATION
MT
M3
M2
BPSG
PECVD
SiN4
Ti/Al
GaN
GaN Buffer Stack
AlN (Seed Layer)
N-epi (4.5um)
Si (100) Substrate
PL
PW
NW
N-epi
NBL
PBL
MP
DT
sub (100)
102
104
106
108
110
120
130
140
152
154
156
157
158
160
162
164a
164b
164c
166a
166b
168a
168b
168c
172
174
176
180
182
184
186
188
190
192

FIG. 1

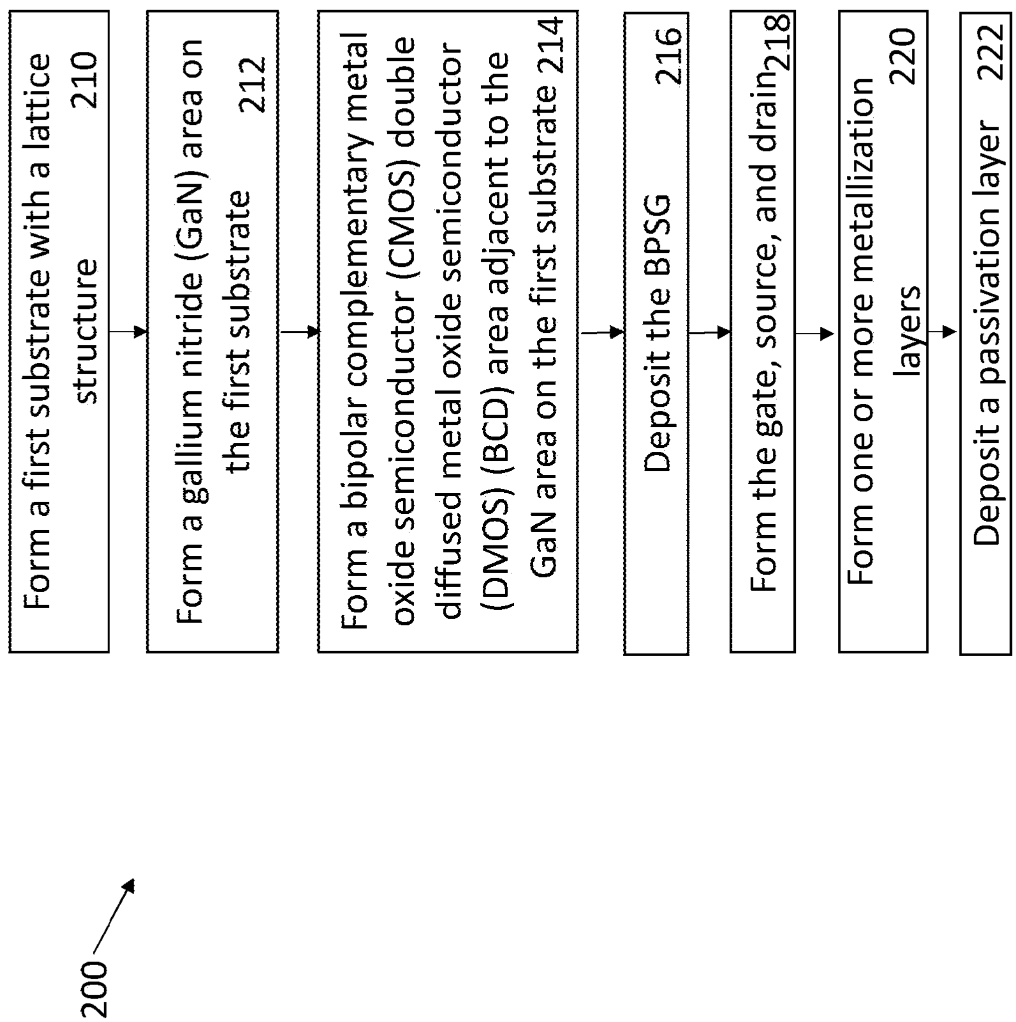
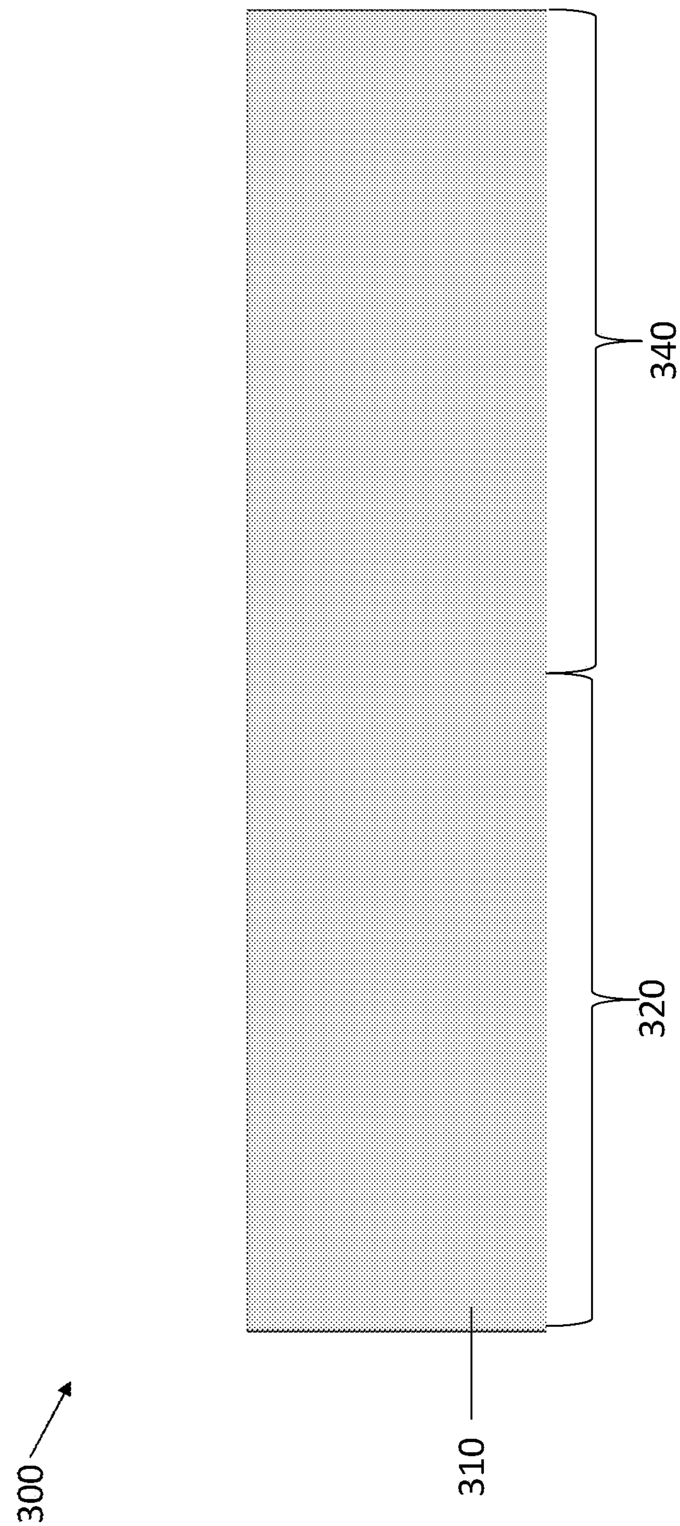

Form a first substrate with a lattice structure 210
Form a gallium nitride (GaN) area on the first substrate 212
Form a bipolar complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area adjacent to the GaN area on the first substrate 214
Deposit the BPSG 216
Form the gate, source, and drain 218
Form one or more metallization layers 220
Deposit a passivation layer 222
200

FIG. 2

FIG. 3A



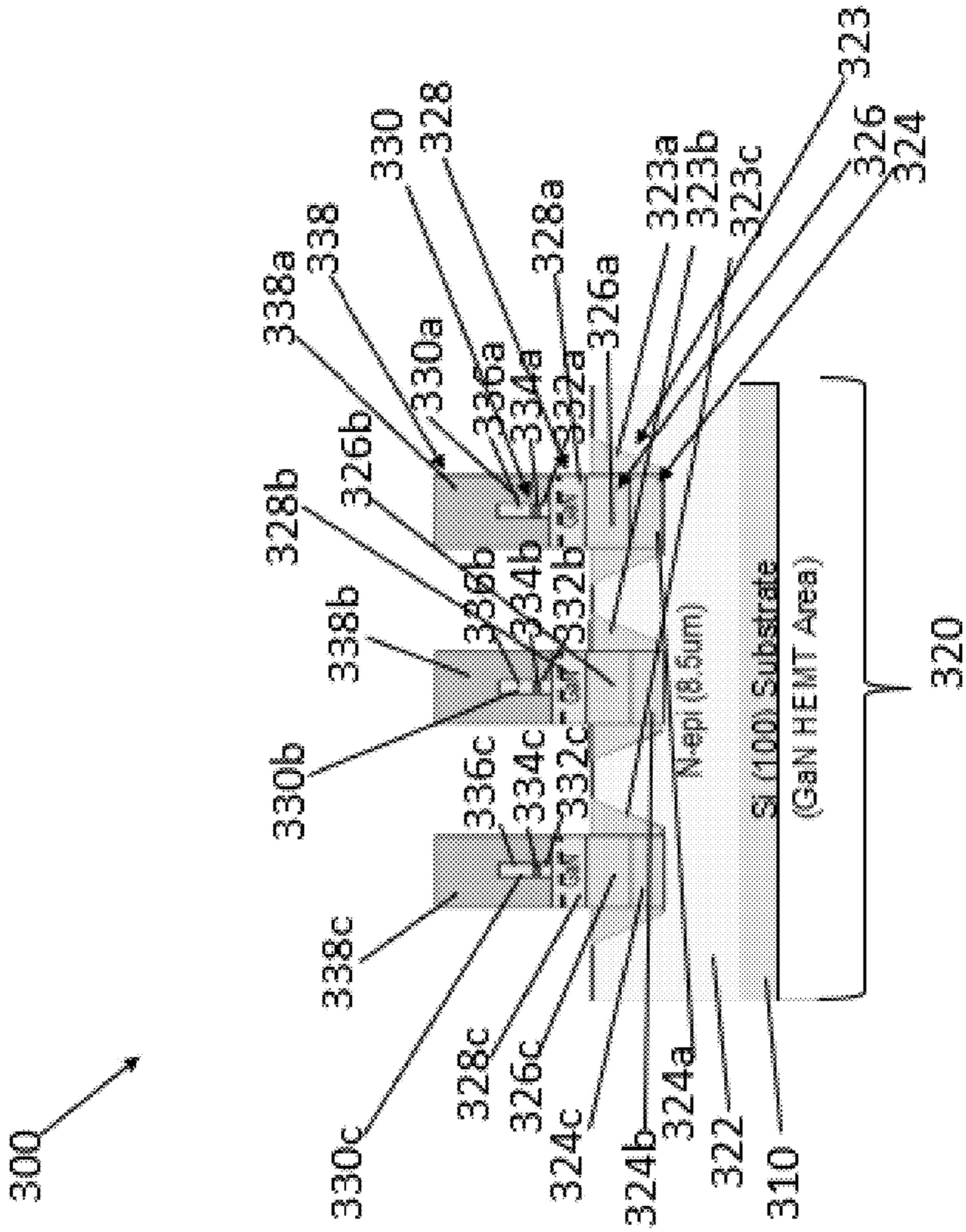
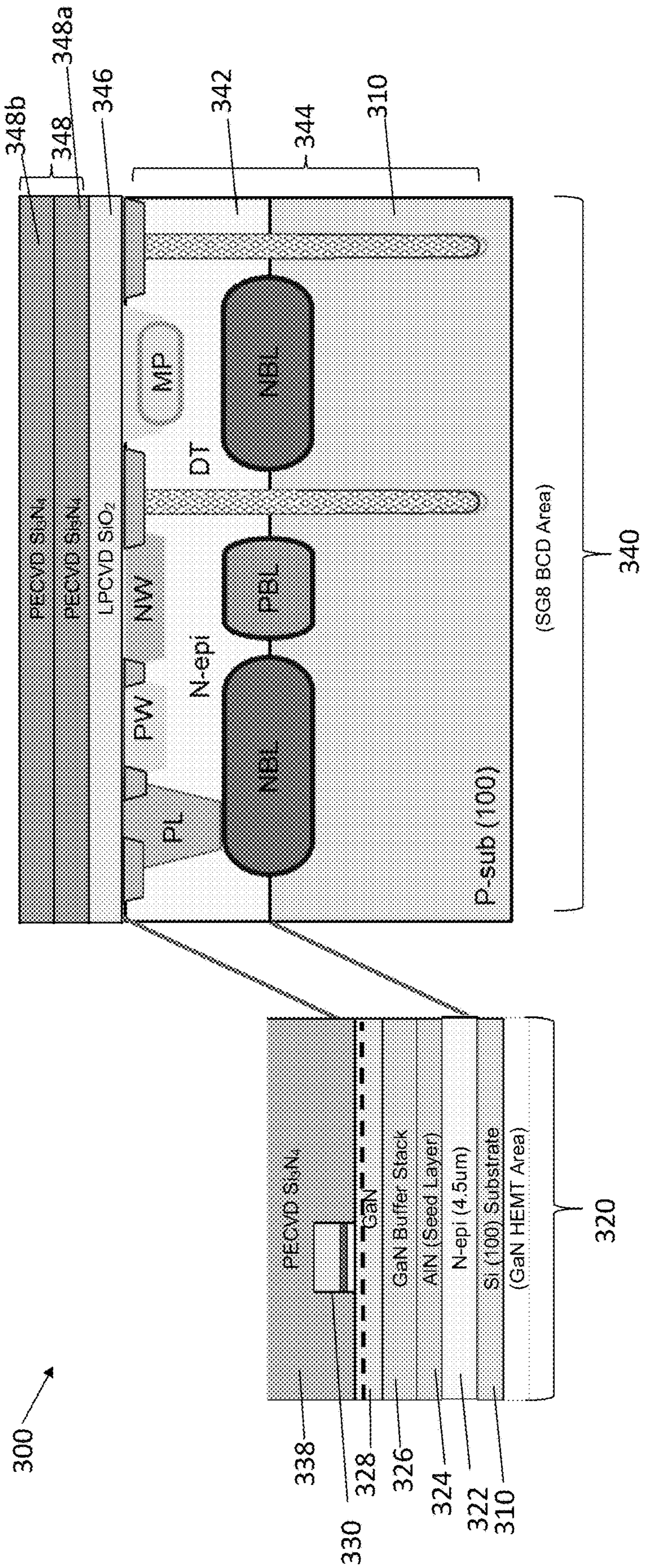

300
338a
338
330
328
328a
330a
336a
334a
332a
326a
323a
323b
323c
323
326
324
326b
328b
336b
334b
332b
338b
330b
336c
334c
332c
N-epi (8.5um)
Si (100) Substrate
(GaN HEMT Area)
320
338c
330c
328c
326c
324c
324b
324a
322
310

FIG. 3B

FIG. 3C


300
348b
348
348a
346
342
344
310
PECVD Si_3N_4
PECVD Si_3N_4
LPCVD SiO_2
MP
NBL
DT
NW
PBL
N-epi
PW
NBL
PL
P-sub (100)
(SG8 BCD Area)
340
PECVD Si_3N_4
GaN
GaN Buffer Stack
AlN (Seed Layer)
N-epi (4.5um)
Si (100) Substrate
(GaN HEMT Area)
320
338
330
328
326
324
322
310

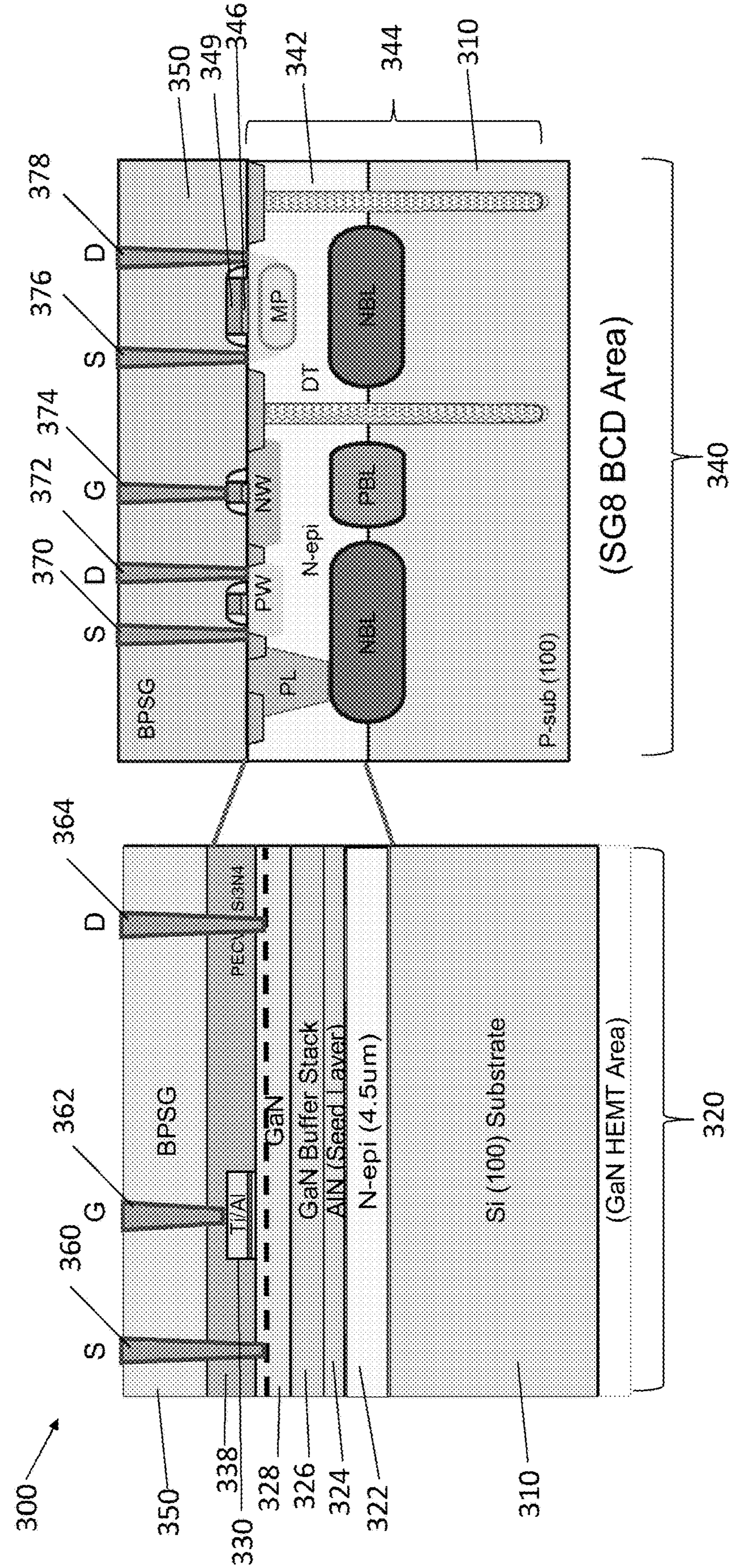
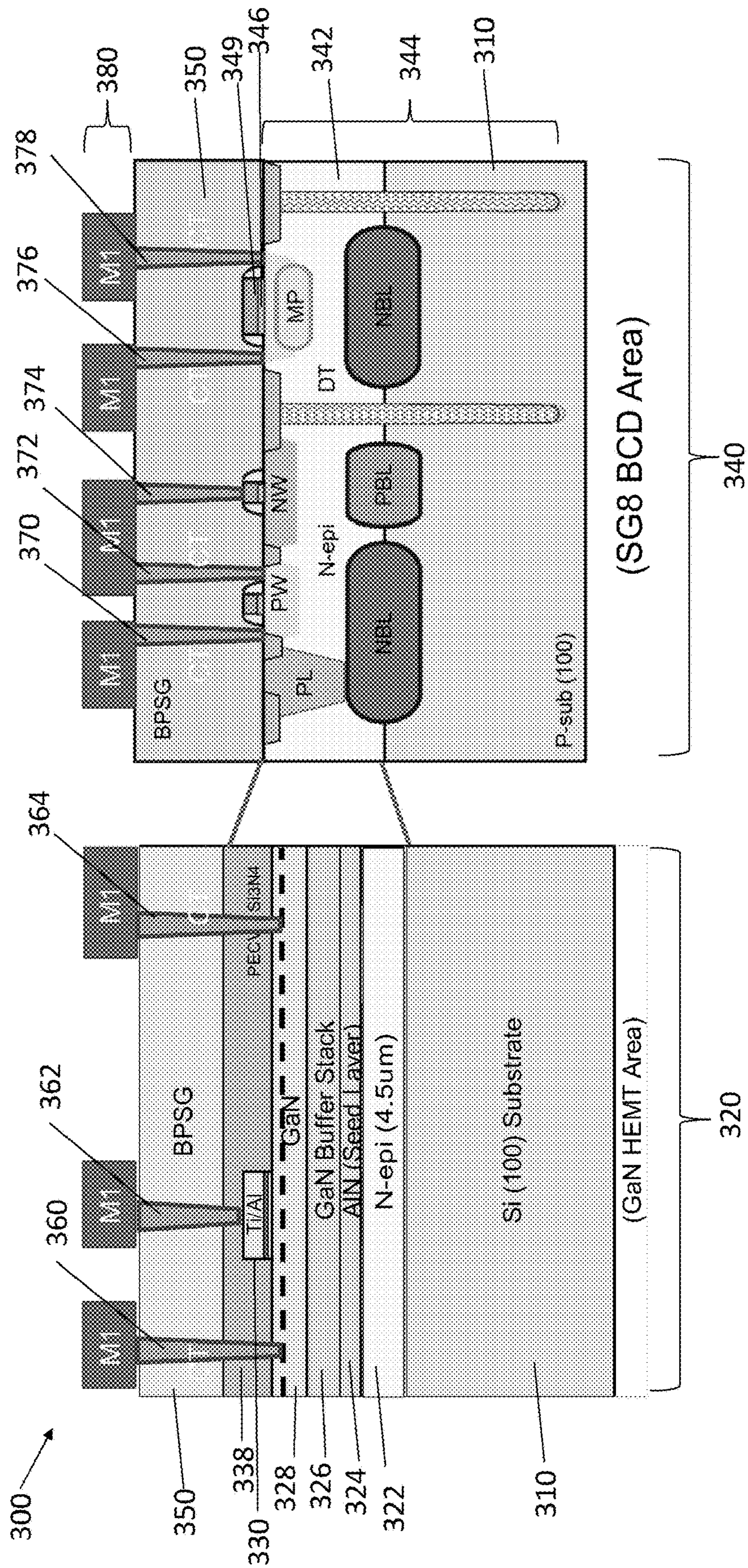

300
S
G
D
360
362
364
BPSG
Ti/Al
PEC
Si3N4
GaN
GaN Buffer Stack
AlN (Seed Layer)
N-epi (4.5um)
Si (100) Substrate
(GaN HEMT Area)
320
350
338
330
328
326
324
322
310
S
D
G
S
D
370
372
374
376
378
BPSG
PL
PW
NW
N-epi
NBL
PBL
DT
MP
P-sub (100)
350
349
346
342
344
310
(SG8 BCD Area)
340

FIG. 3D

FIG. 3E



300
350
338
330
328
326
324
322
310
360
362
364
M1
BPSG
Ti/Al
PECV Si3N4
GaN
GaN Buffer Stack
AlN (Seed Layer)
N-epi (4.5um)
Si (100) Substrate
(GaN HEMT Area)
320
370
372
374
376
378
380
350
349
346
342
344
310
PL
PW
NW
N-epi
NBL
PBL
DT
MP
P-sub (100)
(SG8 BCD Area)
340

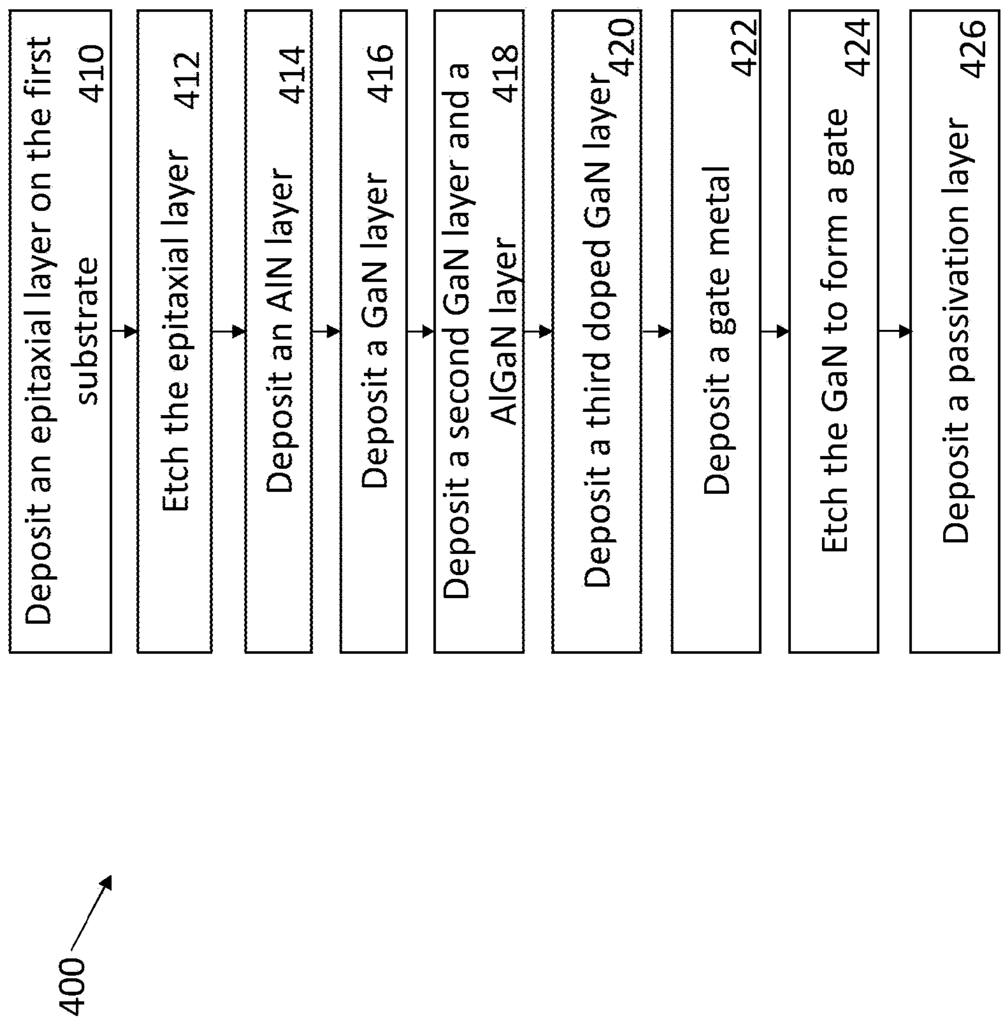
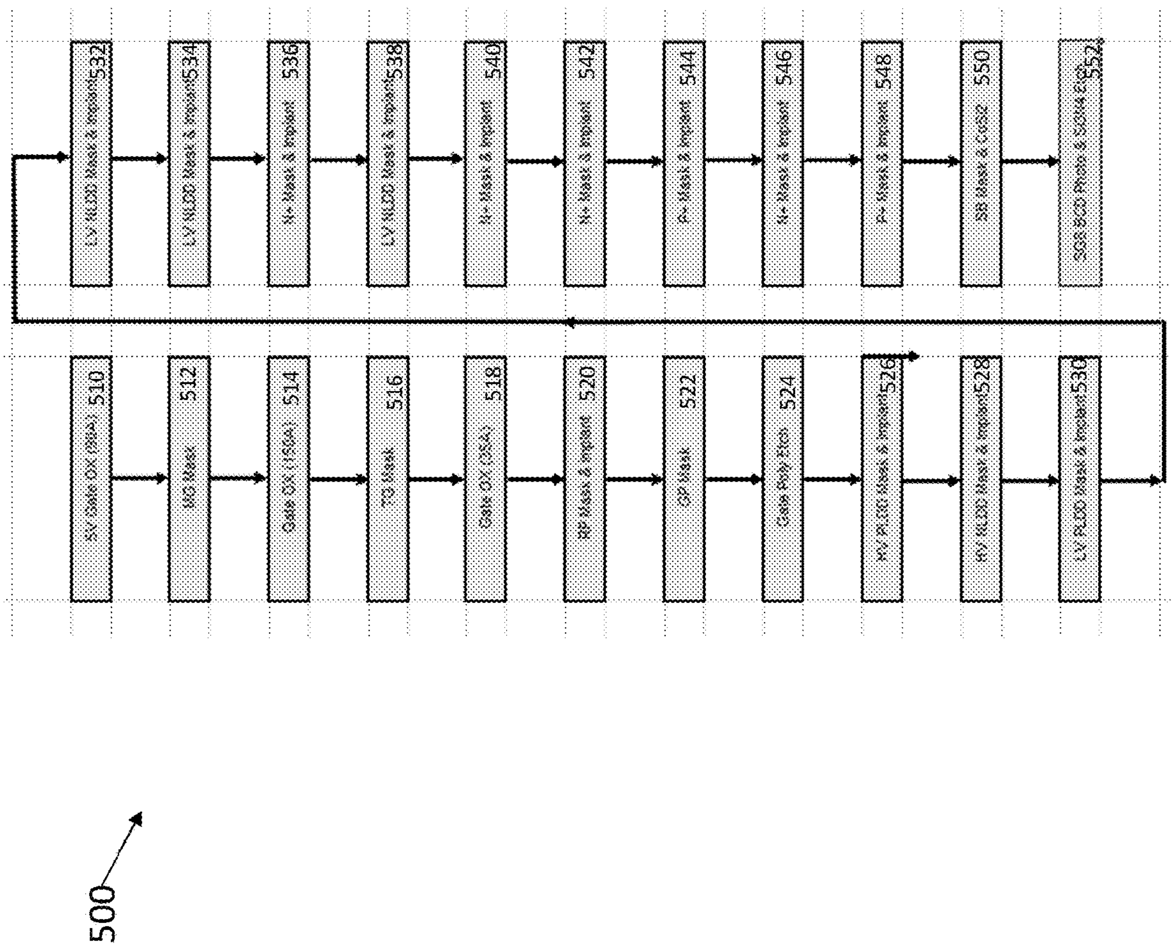

400
Deposit an epitaxial layer on the first substrate 410
Etch the epitaxial layer 412
Deposit an AlN layer 414
Deposit a GaN layer 416
Deposit a second GaN layer and a AlGaN layer 418
Deposit a third doped GaN layer 420
Deposit a gate metal 422
Etch the GaN to form a gate 424
Deposit a passivation layer 426

FIG. 4


500
510
512
514
516
518
520
522
524
526
528
530
532
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536
538
540
542
544
546
548
550
552

FIG. 5

HYBRID GAN AND BCD DEVICES USING HETEROEPITAXY ON SILICON

BACKGROUND

[0001] A bipolar junction transistor (which may herein be referred to as a BJT or bipolar) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) device combines the strengths of three technologies onto a single integrated circuit (IC). A BJT enables high precision analog signal control. A low voltage CMOS transistor is commonly used in applications for high density digital circuits. A DMOS transistor is a type of power transistor commonly used in high voltage applications. The BCD device maybe disposed on a silicon wafer.

[0002] Some semiconductor devices utilize non-homogeneous semiconductor layers. Device layers may include one or more elements, such as gallium nitride (GaN) layers, aluminum nitride (AlN) layers, and aluminum gallium nitride (AlGaN) layers. The non-homogeneous semiconductor layers maybe disposed on a silicon wafer.

SUMMARY

[0003] According to one aspect of the present disclosure, a semiconductor device includes a first substrate having a lattice structure, wherein the first substrate includes a gallium nitride (GaN) area adjacent to a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area. In some embodiments, the GaN area comprises one or more GaN device layers disposed on the first substrate. In some embodiments, the BCD area comprises one or more BCD device layers. In some embodiments, the first substrate comprises a silicon (100) lattice structure configuration.

[0004] According to one aspect of the present disclosure, a method for forming a semiconductor device includes forming a first substrate having a lattice structure. In some embodiments, the method includes forming a gallium nitride (GaN) area on the first substrate. In some embodiments, the method includes forming a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area adjacent to the GaN area on the first substrate. In some embodiments, forming the GaN area comprises depositing one or more GaN device layers on the first substrate. In some embodiments, forming the BCD area comprises depositing one or more BCD device layers. In some embodiments, the first substrate comprises a silicon (100) lattice structure configuration.

DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0005] The manner and process of making and using the disclosed embodiments may be appreciated by reference to the figures of the accompanying drawings. It should be appreciated that the components and structures illustrated in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principals of the concepts described herein. Like reference numerals designate corresponding parts throughout the different views. Furthermore, embodiments are illustrated by way of example and not limitation in the figures, in which:

[0006] FIG. **1** is a side view of a cross section of a semiconductor device having a gallium nitride (GaN) area adjacent to a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area;

[0007] FIG. **2** is a flowchart of an example of a process to fabricate the semiconductor device of FIG. **1**;

[0008] FIG. **3**A is a side view of a cross section of a semiconductor device having a substrate;

[0009] FIG. **3**B is a side view of a cross section of a semiconductor device having a substrate with the GaN area formed on the substrate;

[0010] FIG. **3**C is a side view of a cross section of a semiconductor device having the BCD area formed adjacent to the GaN area on the substrate;

[0011] FIG. **3**D is a side view of a cross section of a semiconductor device having a borophosphosilicate glass (BPSG) layer deposited, and one or more gate contacts, source contacts, and drain contacts formed;

[0012] FIG. **3**E is a side view of a cross section of a semiconductor device having metallization layer deposited on the BPSG layer, gate contacts, source contacts, and drain contacts;

[0013] FIG. **4** is a flowchart of an example of a process to fabricate the GaN area for the semiconductor device of FIG. **1**; and

[0014] FIG. **5** is a flowchart of an example of a process to fabricate the BCD area for the semiconductor device of FIG. **1**.

DETAILED DESCRIPTION

[0015] FIG. **1** shows a semiconductor device **100** comprising a substrate **102** having a gallium nitride (GaN) area **110** adjacent to a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area **120**. One or more metallization layers **130** are disposed on the GaN area **110** and the BCD area **120**. Disposed on the metallization layers **130** is a passivation layer **140**.

[0016] In example embodiments, the substrate **102** is formed from a silicon (100) wafer formed from a single crystal with a lattice structure that is defined by the orientation. Orientation refers to the growth plane of the crystalline silicon. Different crystallographic planes and directions have different arrangements of atoms or lattices when viewed by a particular angle. The different views of the crystallographic orientation of the surface of the wafer may be referred to as silicon (100), silicon (110), and silicon (111). The orientation of these planes and directions can affect the properties of the wafer and resulting device fabricated atop it.

[0017] Silicon (100) refers to the crystallographic plane and direction of the crystal lattice of the silicon wafer. The (100) direction is formed along a line parallel to the y-axis of the crystal lattice. The crystal plane of the silicon (100) is perpendicular to the y- axis of the crystal lattice. Silicon (100) may be referred to herein as a cubic or diamond cubic lattice structure. It is understood that the substrate **102** is one substrate with one lattice structure, but may be referred to herein as a GaN area substrate and a BCD area substrate to facilitate an understanding of the disclosure. The substrate **102** may be doped with a p-type dopant with a resistivity of 5 ohm-cm.

[0018] In the illustrated embodiment, one or more deep trench structures **104**, **106**, **108** isolate the GaN area **110** and the BCD area **120**. A first deep trench structure **104** and a second deep trench structure **106** are positioned on opposing sides of the GaN area **110**. A third deep trench structure **108** is positioned adjacent to the second deep trench structure **106**, in the BCD area **120**. The deep trench structures isolate the voltages and temperatures of the respective GaN area **110** and the BCD area **120**. Multiple deep trench structures may be used to further isolate the two areas. It will be appreciated by those of ordinary skill in the art, that although three deep trenches **104**, **106**, **108** are shown other numbers of deep trenches are possible including more or fewer deep trenches.

[0019] The GaN area **110** comprises an epitaxial layer **152** disposed on the substrate **102**. The epitaxial layer **152** may be doped with an n-type dopant to form an n-type epitaxial layer. One or more GaN device layers **154**, **156**, **157** are disposed on the epitaxial layer **152**. The GaN device layers **154**, **156**, **157** comprise one or more layers having a cubic structure and one or more layers having a wurtzite structure. In embodiments, GaN device layer **154** comprises an aluminum nitride (AlN) layer disposed on the epitaxial layer **152**. GaN device layer **156** is disposed on the GaN device layer **154**. GaN device layer **157** is disposed on the GaN device layer **156**.

[0020] Disposed on the GaN device layers **154**, **156**, **157** is a structure **158** formed out of an aluminum gallium nitride (AlGaN) layer, a GaN layer, and a gate metal layer. Disposed on GaN device layer **157** and the structure **158** is a passivation layer **160**. The structure **158** is a p-GaN that is heavily doped by Magnesium (Mg) to form p-type area. This p-GaN depletes 2-Dimensional Electron Gas (2DEG) under the gate so that enables Enhanced Mode of GaN device. Disposed on the passivation layer **160** is a borophosphosilicate glass (BPSG) layer **162**.

[0021] In embodiments, the BCD area **120** of the substrate **102** may be doped with a p-type dopant in order to form a p-type layers like p-type well (PW), medium deep implanted p-type well (MP), or p+buried layer (PBL). The BCD area **120** comprises an epitaxial layer **174** disposed on the substrate **102**. The epitaxial layer **174** may be doped with an n-type dopant in order to form an n-type layers like n-type well (NW), plug layer implant (PL), or n+buried layer (NBL). One or more bipolar junction transistor (BJT), complementary metal oxide semiconductor (CMOS), and double diffused metal oxide semiconductor (DMOS) (BCD) elements **172** are disposed within the BCD area **120** of the substrate **102** and the epitaxial layer **174**. A gate polysilicon layer **176** is disposed on the BCD elements **172**. The BPSG layer **162** is disposed on the passivation layer **160** and the BCD elements **172**.

[0022] One or more contacts **164***a*, **164***b*, **164***c*, **166***a*, **166***b*, **168***a*, **168***b*, **168***c* are formed within the BPSG layer **162**. The contacts (which may be referred to herein as "vias") extend from the metallization layers **130** to provide conductive paths from the source, gate, and drain, for example. A first source contact **164***a*, a first gate contact **166***a*, and a first drain contact **168***a* are formed in the BPSG layer **162** and the passivation layer **160** in the GaN area **110**. A second source contact **164***b*, a third source contact **164***c*, a second gate contact **166***b*, a second drain contact **168***b*, and a third drain contact **168***c* are formed in the BPSG layer **162** in the BCD area **120**. The contacts **164***a*, **164***b*, **164***c*, **166***a*, **166***b*, **168***a*, **168***b*, **168***c* provide conductive paths from the metallization layers **130** to the device layer elements (such as the GaN device layers **154**, **156**, **157**, and BCD elements **172**).

[0023] In embodiments, the metallization layers **130** are formed above the GaN area **110** and the BCD area **120** of the substrate **102**. In the illustrated embodiment, the metallization layers **130** comprise a first metal layer **180** over the BPSG layer **162**. A first metallization layer **182** is disposed on the first metal layer **180**. A second metal layer **184** is disposed on the first metallization layer **182**. A second metallization layer **186** is disposed on the second metal layer **184**. A third metal layer **188** is disposed on the second metallization layer **186**. A third metallization layer **190** is disposed on the third metal layer **188**. A fourth metal layer **192** is disposed on the third metallization layer **190**. The passivation layer **140** is disposed on the fourth metal layer **192**. It is understood that any practical number of metallization layers, vias, and metal layers can be used to meet the needs of a particular application.

[0024] FIG. **2** shows a flowchart of an example process **200** to fabricate the semiconductor device **100** of FIG. **1**. Process **200** forms a first substrate with a lattice structure in block **210**. Process **200** forms a GaN area on the first substrate in block **212**. Process **200** forms a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area adjacent to the GaN area on the first substrate in block **214**.

[0025] Process **200** deposits the BPSG in block **216**. Process **200** forms the gate contact, source contact, and drain contact in block **218**. Process **200** forms one or more metallization layers in block **220**. Process **200** deposits a passivation layer in block **222**.

[0026] FIGS. **3**A-**3**E describe an example of a process to fabricate the semiconductor device **100** of FIG. **1**, the process is similar to or the same as process **200**. FIG. **3**A shows a semiconductor device **300** includes a substrate **310** having a lattice structure. The substrate **310** may be similar to or the same as substrate **102**. The semiconductor device **300** shown in FIG. **3**A may have been formed using the process described in block **210** in FIG. **2**. Semiconductor device **300** comprises the substrate **310** having a GaN area **320** adjacent to a BCD area **340**. The substrate **310** is one substrate with one lattice structure, but may be discussed herein as a GaN area substrate and a BCD area substrate to facilitate an understanding of the disclosure. The substrate **310** may be doped with a p-type dopant.

[0027] FIG. **3**B shows the GaN area **320** formed on the GaN area **320** of the substrate **310**. The GaN area **320** may be similar to or the same as GaN area **110**. The semiconductor device **300** shown in FIG. **3**B may have been formed using the process described in block **212** in FIG. **2**. The GaN area **320** comprises an epitaxial layer **322** disposed on the substrate **310**. The epitaxial layer **322** may be doped with an n-type dopant in order to form an n-type epitaxial layer.

[0028] One or more trenches **323** are disposed in the epitaxial layer **322**. A first trench **323***a* is disposed adjacent to a second trench **323***b*, a third trench **323***c* is disposed adjacent to the second trench **323***b*. It will be appreciated by those of ordinary skill in the art, that although three trenches **323***a*, **323***b*, **323***c* are shown, other numbers of trenches are possible including more or fewer trenches.

[0029] One or more GaN device layers **324**, **326**, **328** are disposed on the epitaxial layer **322** in the trenches **323**. A GaN device layer **324** comprises aluminum nitride (AlN) and is disposed on the epitaxial layer **322** in the trenches **323**. A first GaN device layer **324***a* is disposed in the first trench **323***a*, a second GaN layer **324***b* is disposed in the second trench **323***b*, and a third GaN layer **324***c* is disposed in the third trench **323***c*. A GaN layer **326** is disposed on the GaN device layer **324**. A first GaN layer **326***a* is disposed on the first GaN device layer **324***a*, a second GaN layer **326***b* is disposed on the second GaN device layer **324***b*, and a third GaN layer **326***c* is disposed on the third GaN device layer **324***c*. A GaN layer **328** is disposed on the GaN layer **326**. GaN layer **328** may be referred to herein as a GaN epitaxial layer. A first GaN layer **328***a* is disposed on the first GaN layer **326***a*, a second GaN layer **328***b* is disposed on the second GaN layer **326***b*, a third GaN layer **328***c* is disposed on the third GaN layer **326***c*.

[0030] On the GaN layer **328** is a structure **330** formed out of a p-GaN layer, an aluminum gallium nitride (AlGaN) layer, a GaN layer, and a gate metal layer. A first structure **330***a* is disposed on the first GaN layer **328***a*, a second structure **330***b* is disposed on the second GaN layer **328***b*, and a third structure **330***c* is disposed on the third GaN layer **328***c*. The AlGaN layer is disposed on the GaN layer **328**. A first AlGaN layer **332***a* is disposed on the first GaN layer **328***a*, a second AlGaN layer **332***b* is disposed on the second GaN layer **328***b*, and a third AlGaN layer **332***c* is disposed on the third GaN layer **328***c*. The p-GaN layer is disposed on the AlGaN layer. A first p-GaN layer **334***a* is disposed on the first AlGaN layer **332***a*, a second p-GaN layer **334***b* is disposed on the second AlGaN layer **332***b*, and a third p-GaN layer **334***c* is disposed on the third AlGaN layer **332***c*. The p-GaN layer may be doped with a p-type dopant to form a p-type GaN layer. The gate metal layer is disposed on the GaN layer. A first gate metal layer **336***a* is disposed on the first GaN layer **334***a*, a second gate metal layer **336***b* is disposed on the second GaN layer **334***b*, and a third gate metal layer **336***c* is disposed on the third GaN layer **334***c*. The gate metal layer **336** comprises one or more of Titanium (Ti), Titanium Nitride (TiN), Aluminum Copper (AlCu). For example, the gate metal **336** may comprise a mixture of Ti/TiN/AlCu/Ti/TiN.

[0031] A passivation layer **338** is disposed on the GaN layer **328** and the structure **330**. A first passivation layer **338***a* is disposed on the first GaN layer **328***a* and the first structure **330***a*, a second passivation layer **338***b* is disposed on the second GaN layer **328***b* and the second structure **330***b*, and a third passivation layer **338***c* is disposed on the third GaN layer **328***c* and the third structure **330***c*. The passivation layer **338** may be formed from a silicon nitride (Si_3N_4) and disposed through a plasma-enhanced chemical vapor deposition process.

[0032] FIG. **3**C shows the BCD area **340** formed adjacent to the GaN area **320** on the substrate **310**. The BCD area **340** may be similar to or the same as BCD area **120**. The semiconductor device **300** shown in FIG. **3**C may have been formed using the process described in block **214** in FIG. **2**. The BCD area **340** of the substrate **310** may be doped with a p-type dopant in order to form a p-type substrate.

[0033] An epitaxial layer **342** is disposed on the BCD area **340** of the substrate **310**. The epitaxial layer **342** may be doped with an n-type dopant in order to form an n-type epitaxial layer. One or more bipolar junction transistor (BJT or bipolar), complementary metal oxide semiconductor (CMOS), and double diffused metal oxide semiconductor (DMOS) (BCD) elements **344** are disposed within the BCD area substrate **314** and the epitaxial layer **342**.

[0034] A silicon dioxide (SiO_2) layer **346** is disposed on the BCD elements **344** and the epitaxial layer **342**. The SiO_2 layer **346** may be disposed through a low pressure chemical vapor deposition process. One or more silicon nitride layers **348** are disposed on the SiO_2 layer **346**. A first silicon nitride layer **348***a* is disposed on the SiO_2 layer **346**. A second silicon nitride layer **348***b* is disposed on the first silicon nitride layer **348***a*. The silicon nitride layers **348** may be disposed through a plasma-enhanced chemical vapor deposition process.

[0035] FIG. **3**D shows the silicon nitride layers **348** etched away, a BPSG layer **350** deposited, and gate contact, source contact, and drain contact formed. The semiconductor device **300** shown in FIG. **3**D may have been formed using the process described in blocks **216**, **218** in FIG. **2**. In the GaN area **320**, the BPSG layer **350** is disposed on the passivation layer **338**. A first source contact **360**, a first gate contact **362**, and a first drain contact **364** are formed in the BPSG layer **350** and the passivation layer **338** in the GaN area **320**. In the BCD area **340**, the BPSG layer **350** is deposited on the gate polysilicon layer **349**, SiO_2 layer **346**, and the BCD elements **344**. A second source contact **370**, a second gate contact **374**, a third source contact **376**, a second drain contact **372**, and a third drain contact **378** are formed in the BPSG layer **350** in the BCD area **340**.

[0036] FIG. **3**E shows metallization layers disposed on the contacts **360**, **362**, **364**, **370**, **372**, **374**, **376**, **378** and the BPSG layer **350**. The semiconductor device **300** shown in FIG. **3**E may have been formed using the process described in blocks **220** in FIG. **2**. The metallization layers may be similar to or the same as the metallization layers **130** in FIG. **1**. The contacts **360**, **362**, **364**, **370**, **372**, **374**, **376**, **378** provide conductive paths from the metallization layers to the device layer elements (such as the GaN device layers **324**, **326**, **328**, and BCD elements **344**). A metal layer **380** is disposed on the BPSG layer **350**, the source contacts **360**, **370**, **376**, the drain contacts **364**, **378**, and the gate contacts **362**, **374**. The metal layer **380** comprises one or more metal layers deposited on the GaN area **320** and the BCD area **340**.

[0037] FIG. **4** is an example of a process to form a GaN area for a semiconductor device, is process **400**. For example, process **400** may be used to form the GaN area **110** of semiconductor device **100**. Process **400** deposits an epitaxial layer on the first substrate in block **410**. The initial epitaxial layer is about 8.5 μm (+/−0.5 μm) thickness.

[0038] Process **400** etches the epitaxial layer in block **412**. The initial epitaxial layer is etched to 4.5 μm (+/−0.5 μm) to form one or more GaN trenches, which levels off the GaN and BCD areas after the deposition of the GaN layers. The etch may be a plasma etch. Following, the etch forms one or more GaN trenches in the epitaxial layer and the semiconductor device is annealed. The GaN trenches are about 4 μm to 5 μm in depth (+/−0.5 μm).

[0039] Process **400** deposits an AlN layer in block **414**. The deposition may include a seed layer growth nucleation of about 1 nm (+/−0.1 nm).

[0040] Process **400** deposits a first GaN layer in block **416**. The GaN layer has a thickness of about 4 μm (+/−0.2 μm). The first GaN layer may be deposited through a metalorganic chemical vapor deposition (MOCVD) process with a growth temperature of about 900° C. to 1100° C. (+/−10° C.).

[0041] Process **400** deposits a second GaN layer and a AlGaN layer in block **418**. The AlGaN layer may have a thickness of about 0.7 nm (+/−0.1 nm).

[0042] Process **400** deposits a third doped GaN layer in block **420**. The third doped GaN may have a thickness of about 85 nm (+/−5 nm). The third doped GaN layer may be doped with a p-type dopant to form a p-type GaN layer. The p-type dopant may be magnesium (Mg) with a doping level of 6×10^{19} or 4×10^{19}.

[0043] Process **400** deposits a gate metal in block **422**. The gate metal may comprise a mixture of Ti/TiN/AlCu/Ti/TiN. The gate metal may form an ohmic contact. High temperature annealing may be used to activate the dopants in the third doped GaN layer.

[0044] Process **400** etches the GaN to form a gate in block **424**. The process **400** etches down to gate metal and p-GaN to form a gate.

[0045] Process **400** deposits a passivation layer in block **426**. The passivation layer may comprise Si_3N_4 and may be deposited through a plasma-enhanced chemical vapor deposition process.

[0046] Referring now to FIG. **5**, an example of a process to form a BCD area for a semiconductor device is process **500**. For example, process **500** may be used to form the BCD area **120** of semiconductor device **100**. Process **500** deposits a gate oxide in block **510**. The gate oxide may be a 5V gate oxide (90 angstrom). Process **500** deposits a mask in block **512**. The mask may be an medium gate oxide (MG) mask. Process **500** forms a gate oxide in block **514**. The gate oxide may be a gate oxide (**150** angstrom). Process **500** deposits a phosphorus implant (RP) mask and implant in block **520**. Process **500** deposits a gate polysilicon (GP) mask in block **522**.

[0047] Process **500** etches the gate polysilicon in block **524**. Process **500** deposits a high voltage (HV) p-type implantation of the lightly doped diffusion (PLDD) mask and implant in block **526**. Process **500** deposits a HV n-type implantation of the lightly doped diffusion (NLDD) mask and implant in block **528**. Process **500** deposits a low voltage (LV) PLDD mask and implant in block **530**. Process **500** deposits a LV NLDD mask and implant in block **532**. Process **500** deposits a LV NLDD mask and implant in block **534**. Process **500** deposits a N+ mask and implant in block **536**. Process **500** deposits a LV NLDD mask and implant in block **538**. Process **500** deposits a N+ mask and implant in block **540**. Process **500** deposits a N+ mask and implant in block **542**. Process **500** deposits a P+mask and implant in block **544**. Process **500** deposits a N+ mask and implant in block **546**. Process **500** deposits a P+ mask and implant in block **548**. Process **500** deposits a SB mask and cobalt silicide ($CoSi_2$) in block **550**.

[0048] Process **500** deposits a photomask and etches a passivation layer in block **552**.

[0049] The BCD process delivers power integrated circuit designs with an operating voltage of up to 120V. This operating voltage enables exceptional energy efficiency and integration that combines analog circuits and digital content as well as power devices and embedded non-volatile memories (NVMs). Integration of the BCD area with the GaN area on a single substrate offers a reduction of power loss, reduction of system size, improvement of efficiency, and a reduction of cost.

[0050] Although reference is made herein to particular materials, it is appreciated that other materials having similar functional and/or structural properties may be substituted where appropriate, and that a person having ordinary skill in the art would understand how to select such materials and incorporate them into embodiments of the concepts, techniques, and structures set forth herein without deviating from the scope of those teachings.

[0051] Various embodiments of the concepts, systems, devices, structures and techniques sought to be protected are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of the concepts, systems, devices, structures and techniques described herein. It is noted that various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the described concepts, systems, devices, structures and techniques are not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship.

[0052] As an example of an indirect positional relationship, references in the present description to forming layer "A" over layer "B" include situations in which one or more intermediate layers (e.g., layer "C") is between layer "A" and layer "B" as long as the relevant characteristics and functionalities of layer "A" and layer "B" are not substantially changed by the intermediate layer(s). The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms "comprises," "comprising, "includes," "including," "has," "having," "contains" or "containing," or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

[0053] Additionally, the term "exemplary" is used herein to mean "serving as an example, instance, or illustration. Any embodiment or design described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms "one or more" is understood to include any integer number greater than or equal to one, i.e. one, two, three, four, etc. The terms "a plurality" are understood to include any integer number greater than or equal to two, i.e. two, three, four, five, etc. The term "connection" can include an indirect "connection" and a direct "connection."

[0054] References in the specification to "one embodiment, "an embodiment," "an example embodiment," etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0055] For purposes of the description hereinafter, the terms "upper," "lower," "right," "left," "vertical," "horizontal, "top," "bottom," and derivatives thereof shall relate to the described structures and methods, as oriented in the drawing figures. The terms "overlying," "atop," "on top, "positioned on" or "positioned atop" mean that a first element, such as a first structure, is present on a second element, such as a second structure, where intervening elements such as an interface structure can be present between the first element and the second element. The term "direct contact" means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary elements.

[0056] Use of ordinal terms such as "first," "second," "third," etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

[0057] The terms "approximately" and "about" may be used to mean within ±20% of a target value in some embodiments, within ±10% of a target value in some embodiments, within ±5% of a target value in some embodiments, and yet within ±2% of a target value in some embodiments. The terms "approximately" and "about" may include the target value. The term "substantially equal" may be used to refer to values that are within ±20% of one another in some embodiments, within ±10% of one another in some embodiments, within ±5% of one another in some embodiments, and yet within ±2% of one another in some embodiments.

[0058] The term "substantially" may be used to refer to values that are within ±20% of a comparative measure in some embodiments, within ±10% in some embodiments, within ±5% in some embodiments, and yet within ±2% in some embodiments. For example, a first direction that is "substantially" perpendicular to a second direction may refer to a first direction that is within ±20% of making a 90° angle with the second direction in some embodiments, within ±10% of making a 90° angle with the second direction in some embodiments, within ±5% of making a 90° angle with the second direction in some embodiments, and yet within ±2% of making a 90° angle with the second direction in some embodiments.

[0059] It is to be understood that the disclosed subject matter is not limited in its application to the details of construction and to the arrangements of the components set forth in the following description or illustrated in the drawings. The disclosed subject matter is capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the disclosed subject matter. Therefore, the claims should be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the disclosed subject matter.

[0060] Although the disclosed subject matter has been described and illustrated in the foregoing exemplary embodiments, it is understood that the present disclosure has been made only by way of example, and that numerous changes in the details of implementation of the disclosed subject matter may be made without departing from the spirit and scope of the disclosed subject matter.

What is claimed is:

1. A semiconductor device, comprising:

a first substrate having a lattice structure, wherein the first substrate includes a gallium nitride (GaN) area adjacent to a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area,

wherein the GaN area comprises one or more GaN device layers disposed on the first substrate and the BCD area comprises one or more BCD device layers disposed on the first substrate.

2. The semiconductor device of claim **1**, wherein the first substrate comprises a silicon (100) lattice structure configuration.

3. The semiconductor device of claim **1**, wherein the GaN devices layers comprise one or more GaN device layers having a cubic structure and one or more GaN device layers having a wurtzite structure.

4. The semiconductor device of claim **1**, wherein the one or more GaN device layers comprise one or more GaN layers, one or more aluminum nitride (AlN) layers, one or more aluminum gallium nitride (AlGaN) layers, and one or more p-type doped p-GaN layers.

5. The semiconductor device of claim **3**, further comprising an epitaxial layer between the first substrate and a first AlN layer.

6. The semiconductor device of claim **1**, wherein an area of the first substrate having the BCD area further comprises an epitaxial layer with an n-type dopant and an area of the first substrate having the GaN area further comprises an epitaxial layer with an n-type dopant and a GaN epitaxial layer.

7. The semiconductor device of claim **5**, wherein an epitaxial layer with an n-type dopant is disposed on the first substrate, wherein the first substrate has a p-type dopant.

8. A method, comprising:

forming a first substrate having a lattice structure;

forming a gallium nitride (GaN) area on the first substrate; and

forming a bipolar junction transistor (BJT) complementary metal oxide semiconductor (CMOS) double diffused metal oxide semiconductor (DMOS) (BCD) area adjacent to the GaN area on the first substrate,

wherein forming the GaN area comprises depositing one or more GaN device layers on the first substrate and forming the BCD area comprises depositing one or more BCD device layers on the first substrate.

9. The method of claim **8**, wherein the first substrate comprises a silicon (100) lattice structure configuration.

10. The method of claim **8**, wherein the GaN devices layers comprise one or more GaN device layers having a cubic structure and one or more GaN device layers having a wurtzite structure.

11. The method of claim **8**, wherein forming the GaN device layers further comprises:

depositing an epitaxial layer on the first substrate;

etching the epitaxial layer on the first substrate;

depositing an AlN layer;

depositing a first GaN layer;

depositing a second GaN layer and a AlGaN layer;

depositing a third doped GaN layer;

depositing a gate metal;

etching the GaN to form a gate; and

depositing a passivation layer.

12. The method of claim **11**, wherein etching the epitaxial layer on the first substrate further comprises forming one or more trenches in the epitaxial layer on the first substrate.

13. The method of claim **12**, wherein the AlN layer, first GaN layer, and second GaN layer are deposited on the epitaxial layer on the first substrate in the one or more trenches.

14. The method of claim **8**, wherein the first substrate having the BCD area has a p-type dopant.

15. The method of claim **14**, wherein an epitaxial layer with an n-type dopant is disposed on the first substrate with a p-type dopant.

16. The method of claim **8**, further comprising depositing one or more polysilicon layers on the BCD area and etching the one or more polysilicon layers.

17. The method of claim **8**, further comprising depositing a borophosphosilicate glass (BPSG) layer.

18. The method of claim **17**, further comprising forming one or more source contacts, one or more drain contacts, and one or more gate contacts.

19. The method of claim **8**, further comprising depositing one or more metallization layers on the GaN area and the BCD area.

20. The method of claim **19**, further comprising depositing a passivation layer on the one or more metallization layers.

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